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An optimized cloud-based blockchain framework for enhanced scalability, security, and energy efficiency

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Abstract

Blockchain technology functions as an innovative solution that allows users to obtain secure decentralized tamper-resistant data management capabilities. The traditional blockchain systems demonstrate drawbacks because they create scalability challenges and need high amounts of energy for operations together with delayed transaction times, especially when used in high-throughput and resource-constrained situations. This research designs an improved cloud-based blockchain system which combines Proof-of-Stake (PoS) protocol with Byzantine Fault Tolerance (BFT) controls, along with batch transaction handling systems, to foster better performance results. The framework delivers dynamic scalability and distributed processing efficiency because it makes use of cloud infrastructure. The implementation simulation reveals substantial increases in transaction rate upto 10,000 TPS and quicker consensus times together with enhanced validator impartiality as well as more than 60% energy efficiency improvement, as compared to standard Proof-of-Work protocols. Security and trust evaluation methods exhibit thorough testing of the system that confirms its capabilities in adverse environments. This proposed framework provides an energy-efficient solution which can serve robust real-world applications in IoT together with healthcare and supply chain and finance domains.

Keywords Blockchain optimization, Proof-of-Stake (PoS), Byzantine fault tolerance (BFT), Cloud computing, Energy efficiency, Transaction throughput, Batch processing, Validator consensus, Scalable blockchain framework, Secure decentralized systems

1 Introduction

Decentralized trust and secure data integrity in distributed systems now finds its solution with revolutionary blockchain technology. The platform which began serving cryptocurrencies, now acts as the fundamental base for applications in healthcare alongside supply chain management, finance and cloud systems so it evolved quickly [1]. Conventionally used blockchain technologies face substantial drawbacks because they create scalability obstacles alongside high power consumption along with low transaction capacity and delayed operations, while operating under time constraints in environments with limited resources such as cloud networks [2]. The growing implementation



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of cloud-based data infrastructure has led to the emergence of Cloud-Based Blockchain Systems which unite blockchain technology with cloud management systems [3]. The implementation of these systems delivers improved data tracking capabilities together with untouchable audit records and enhanced data handling permissions. Implementing blockchain technology poses multiple difficulties which users must overcome. The implementation of cloud-based applications encounters deployment issues because of delayed consensus processes, inefficient transaction grouping processes and proof-of-work operation requirements which affect system responsiveness [4].

A streamlined and optimized blockchain framework for cloud infrastructures controls three main performance sectors such as scalability with enhanced energy efficiency and strengthened security features. The proposed system applies Proof-of-Stake technology as a minimalist consensus mechanism to remove the considerable energy demands of conventional mining operations [5]. A batch processing system for transactions enhances throughput speed and decreases block creation expenses. The system secures its fault-tolerance through BFT and validator reputation management, which both enhance node identification of malicious or faulty components [6].

The objectives of the paper :

- The research will introduce along with assessing a Proof-of-Stake (PoS)-based consensus mechanism that decreases block validation requirements while enhancing overall consensus speed.
- The system will process transactions in batches to speed up performance while reducing the delay during single transaction executions.
- The system requires BFT-based fault-tolerant mechanisms to detect malicious behavior of faulty or malicious validator nodes within the network.
- A performance analysis of blockchain architecture with and without optimization measures should be conducted using real-time simulation data while showing three performance metrics including block mining speed and validator rate and transaction count within blocks.

Research paper follows this structure. Section I explains why blockchain technology should be used for cloud environments while discussing standard blockchain system limitations that exist due to insufficient scalability and excessive energy consumption and delayed consensus. II provides a literature review of blockchain optimization methods which include Proof-of-Stake consensus models alongside batch processing methods and Byzantine Fault Tolerance (BFT) models. Research gaps discovered within this part serve as the foundation for designing the proposed solution. Section III describes the methodology which outlines the optimized blockchain framework's design including details about PoS-based validator selection protocol alongside batch transaction management systems and fault detection protocols and implementation within a cloud-based testing environment. Section IV presents experimental results that compare optimized versus non-optimized model performances based on four metrics such as transaction processing speed and block processing duration as well as validator distribution patterns and energy utilization statistics. All data is displayed through real-time charts. Section V of this paper presents a summary of the main study results followed by a discussion of the practical value for cloud-based applications and proposed future research areas

concerning smart contracts, cross-chain networking and distributed cloud infrastructure deployment.

2 Literature review

The integration between blockchain and IoT infrastructure with cloud systems remains the core research focus because it solves trust concerns and decentralization and protects data security. The research team of Saha et al. [7] presented a blockchain framework design for post-quantum environments to provide security through decentralized mechanisms which fight against quantum computing threats. The study established principles which became essential for developing quantum-secure blockchain protocols because they will be necessary for sustainable future cloud use. While Xu et al. [8] built BeepTrace as a privacy-preserving blockchain-enabled contact tracing system that showed how quick decentralized system designs can help public health infrastructure by sharing secure data for critical time-sensitive cloud-based applications. Kakkar et al. [9] established an EV charging station scheduling mechanism which utilized blockchain technology and IoT components to find optimized consensus solutions. Through their research they proved that blockchain integrated with qualified scheduling and optimization engines minimizes resource contention and strengthens IoT-cloud system execution. Peng et al. [10] presented blockchain technologies for vehicular IoT (V-IoT) through an overview of consensus scalability and security enhancements which impact distributed cloud systems.

Although SklCoin represents a PoS and collective signature-based consensus protocol by Jaroucheh et al. [11] it achieves scalability and consistency features. The protocol design achieves both strong consistency results while selecting validators efficiently which corresponds to our framework's PoS implementation and detection system. A scoping review on Blockchain-Cloud (BcC) architectures by Ismail et al. established privacy protection together with access control measures and data unalterability as fundamental blockchain drivers for medical implementations according to [12, 13]. Their research establishes the critical need for safe and optimized framework systems that process sensitive data within cloud systems. Algarni et al. [14] established blockchain technology as a basis to develop secure access control mechanisms for IoT systems which demonstrate better trust management and reduced risk points in distributed cloud environments. The implementation of blockchain in e-governance resulted in Hjalmarsson et al. [15] creating an electronic voting system where blockchain maintained transparent and verifiable decision-making functions within distributed networks. The proposed model solves agreement finality issues together with voter identification problems by optimizing validators while minimizing energy usage during consensus.

Peelam et al. [16] developed a solution that integrates quantum blockchain into IoT networks to achieve improved security and defend against quantum threats which is vital for large-scale cloud deployments. Jeong [17] created a blockchain method which incorporates multiple hash chains as a means to decrease integrity failures during IoT data storage in cloud platforms. The technique introduced trustable and consistent data logging while proving that optimized blockchain mechanisms boost system resilience performance. The research demonstrates active work to boost blockchain operations within cloud environments by improving its performance as well as security capabilities. A complete framework which unites PoS-based consensus along with batch processing

and BFT-based fault detection and multi-metric optimization in cloud-native infrastructure has not been established yet. The research takes on the goal of creating a complete optimized blockchain system that gets verified by simulation tests and performance evaluations. Recent research by Mallick et al., has examined blockchain-based IoT and cloud systems that focus on the secure sharing of data, decentralized storage, and optimization of performance. These works emphasize the need to have scalable and energy efficient distributed frameworks, which is in line with the goals of the proposed model [18, 19].

3 Methodology

The proposed blockchain framework resulting from optimization enhances both transaction speed and consensus performance while saving energy due to its deployment in a cloud system. The three central methods include (1) stake-weighted Proof-of-Stake (PoS) consensus system and (2) batched transaction processing and (3) fault-tolerant monitoring which draws from Byzantine Fault Tolerance (BFT) concepts [20].

Figure 1 relates the specific system architecture of the proposed blockchain-based cloud framework as it will include the Batch Processing module, Validator Selection module, BFT Fault Detection module, and Cloud Orchestration layer. The components have a specific role to play in the guaranteeing of scalability, security and energy efficiency. The proof-of-stake algorithm replaces former proof-of-work systems for selecting validators through stake weighting. The selection of validators within PoS depends on the cryptocurrency amount they both own and stay willing to use as collateral [21].

The Fig. 2 shows an architecture which uses cloud integration and optimized blockchain operations for real-world applications to enhance both scalability and reliability as well as performance enhancement. The user-initiated transactions flow into the Transaction Pool as the top layer of the stack. A Stake Manager regularly processes the Transaction Pool to maintain records of both participant tokens and stakes. To pick a validator node the Validator Selector module implements a mathematical model which selects

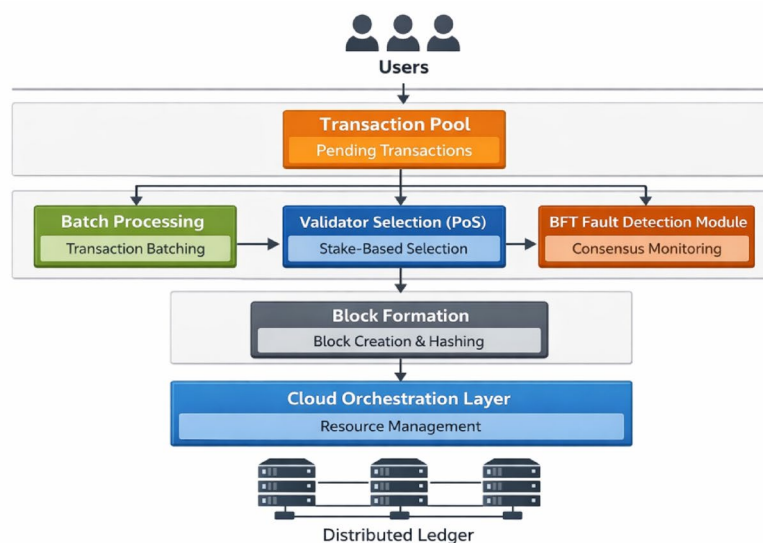


Fig. 1 Proposed architecture diagram

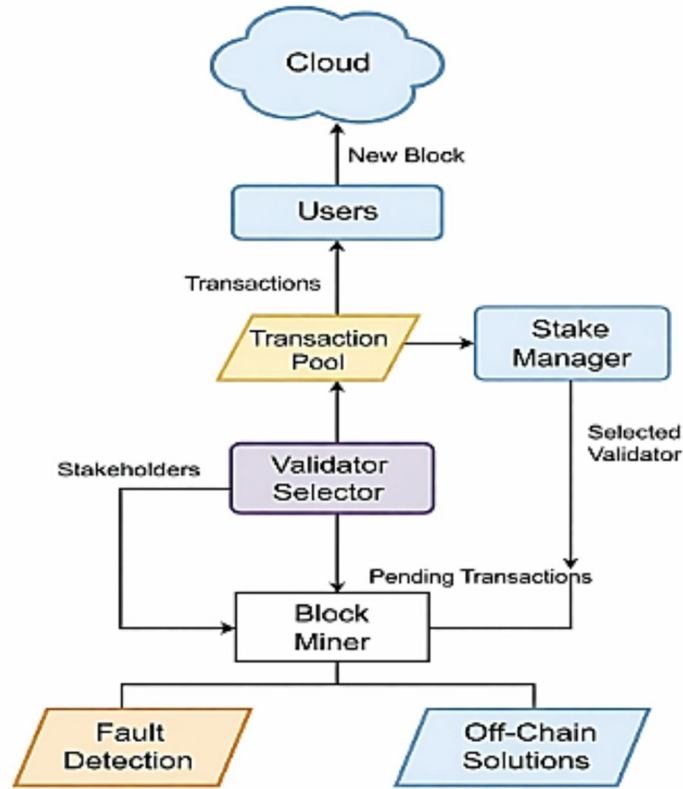


Fig. 2 Optimized cloud-based blockchain architecture

nodes from the Proof-of-Stake (PoS) algorithm on the basis of stake amounts where greater stakes boost selection probabilities. Information about stakeholder identity combined with the selected validator is transmitted to the Block Miner through this process [22]. Judicial operations by the Block Miner include combining open transactions into an upcoming block for addition to the blockchain. The Block Miner integrates with two main subsystems: Fault Detection performs node misbehaving and failure identification via Byzantine Fault Tolerance (BFT) logic and Off-Chain Solutions implements Merkle root validation or sidechains to process high transaction volume and reduce on-chain data duplication. Block creation leads to new block distribution into the cloud infrastructure which offers decentralized storage with increased availability and global purposes [23]. Figure 3 illustrates the optimal blockchain architecture that incorporates PoS, batch processing, and BFT pruning to achieve scalable and efficient consensus.

$$P_i = \frac{S_i}{\sum_{j=1}^n S_j} \quad (1)$$

Let S_j represent the stake of validator j .

The probability P_i of selecting a validator i is defined as:

$$P_i = \frac{S_i}{S_{total}} \quad (2)$$

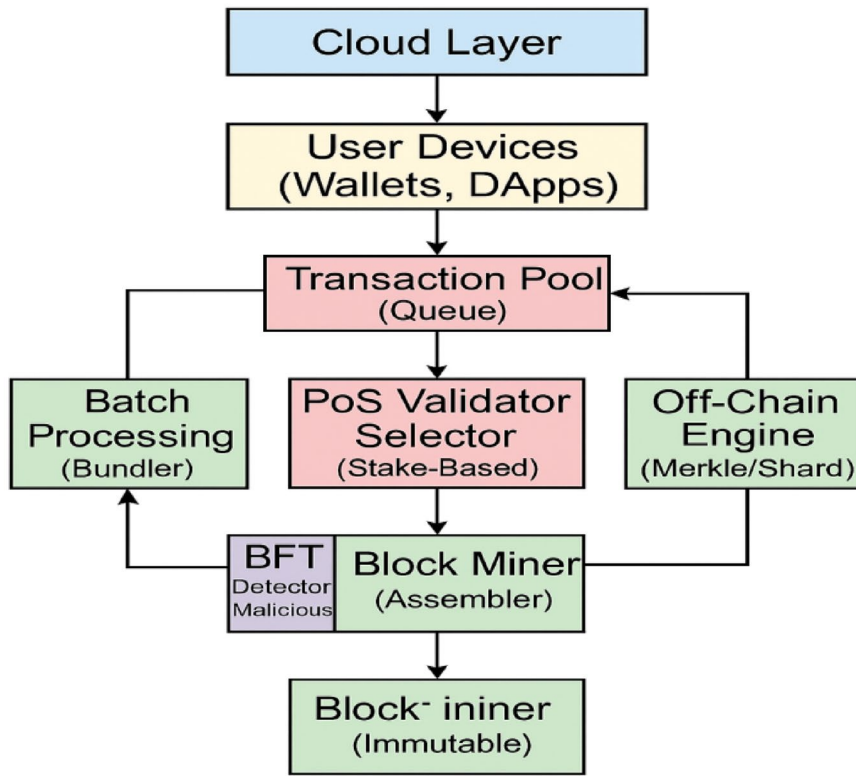


Fig. 3 Optimized blockchain framework for scalability, security, and energy efficiency

Validator fairness is a fairness measure which is defined as the proportionality of stake contribution and block allocation. Preferably, a stakeholder validator.

should be given block assignments based on its proportion of share of the stake. The metric of fairness deviation is the following:

$$F_i = \left| \left(\frac{B_i}{B_{total}} \right) - \left(\frac{S_i}{S_{total}} \right) \right| \quad (3)$$

where:

B_i = blocks assigned to validator I , B_{total} = total blocks, S_i = stake of validator I , S_{total} = total network stake.

The stake-based validation process provides nodes with higher contributions an increased opportunity to validate blocks thereby promoting fairness and minimizing power consumption by doing away with mining challenges. The system improves its performance level through transaction batch processing techniques [24]. Each block gets multiple transactions added to it as a batch instead of just one. The group processing decreases block generation costs while simultaneously boosting transaction speed to . A block mining process takes on average Δ time while the number of processed transactions within each block reaches an average of T .

$$\lambda = \frac{T}{\Delta t} \quad (4)$$

Through the optimized system the block-processing capability at each node can reach higher T values while maintaining stable block frequency. Although the batch processing is better in throughput, to maintain reliability in the face of adversarial scenarios, a fault-tolerant validation scheme should be implemented efficiently, which is met by the proposed BFT-based method.

The block validation and finality guarantee mechanism proposed in the suggested framework is Proof-of-Stake (PoS), which is the main mechanism of the leader selection process. BFT, a lightweight Practical Byzantine Fault Tolerance (PBFT)-inspired protocol, is not implemented as the primary consensus replacement, rather it is applied as a fault-tolerant validation layer, in order to identify malicious or non-responsive validators. The process of selecting the leaders is stake-weighted and the possibility of becoming a leader depends on the contribution of the validator in terms of the stake. In BFT voting rounds, the voting authority is also stake-proportional, so validators with higher stake have more influence in consensus but continue to satisfy Byzantine fault tolerance limits ($f < n/3$) [25]. The implemented BFT mechanism follows a PBFT-inspired three-phase commit model (pre-prepare, prepare, commit) adapted for cloud-based deployment with reduced communication overhead. The Byzantine fault tolerance ($f < n/3$) is imposed on the active set of validators V , such that the pruning mechanism is not forcing consensus safety or correctness to be compromised.

The process excludes from further validator selection to enhance trustworthiness along with system integrity. The value of Research-based trials determine value for reaching performance and resilience balance while Merkle Root calculation defends against blocksize expansion [26]. The system forms a Merkle Tree through recursive pairing and hashing of transactions 1,2,, starting from the individual elements before setting the root R as the fundamental verification mechanism [27].

$$R = H(H(tx1 \parallel tx2) \parallel H(tx3 \parallel tx4) \parallel \dots) \quad (5)$$

Figure 4 presents the Data Flow Diagram depicting the transaction lifecycle from user submission to batch processing, validator selection, BFT consensus verification, block creation, and cloud-based ledger synchronization. The experimental analysis was implemented in a simulation setting running in the cloud that had a total of 1,000 nodes with

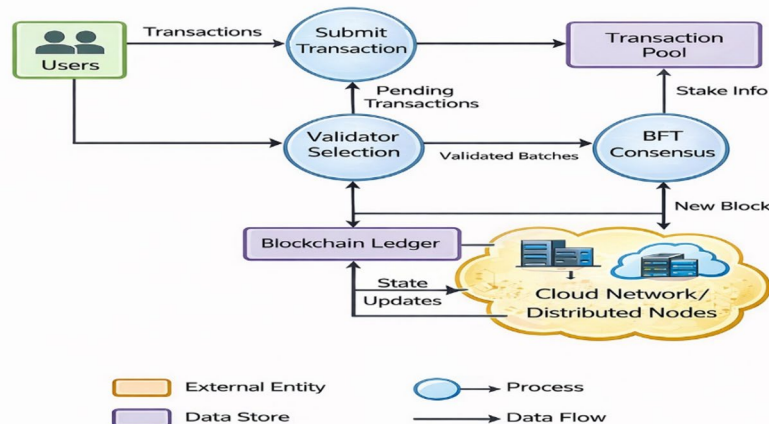


Fig. 4 Data flow diagram

100 nodes as validators. The block-size was set to 2 Mb with average transaction size of 250 bytes, which made around 8,000 transactions per block. The time interval between blocks was adjusted to 2 s. The network assumptions were a minimum bandwidth of 10 Mbps/node and an average propagation latency not greater than 100 ms. Each of the experiments was repeated five times to achieve stability in the results. The experiment was tested on a virtualized environment simulated on a cloud, with the help of virtual machines having 4 vCPUs, 8 GB RAM, and Ubuntu OS. The presented simulation was run on a home-made Python-based framework simulating PoS selection, BFT voting, and updates of trust-score. Results are average of values obtained in several runs with controlled network conditions in order to provide reproducibility. The power consumption of the proposed system was computed with the help of a CPU utilization based energy estimation model based on computational power. Energy consumption was determined by multiplying between the average CPU usage, the time of execution and the power rating per node. The simulated network adheres to a semi-distributed topology with geographically dispersed nodes and managed latency to the environment to represent a realistic deployment of clouds.

The total estimated energy consumption of the blockchain network is computed as:

$$E_{compute} = \sum_{i=1}^N (P_i \times U_i \times T) \quad (6)$$

where:

N = number of active validator nodes.

P_i = rated power of VM instance (W).

U_i = average CPU utilization (0–1).

T = execution time (seconds).

Since the evaluation is conducted in a virtualized cloud environment, P_i represents the nominal power rating corresponding to the allocated VM profile rather than direct physical hardware measurement.

Additionally, communication energy overhead is estimated as:

$$E_{comm} = N \times B \times T \times \gamma \quad (7)$$

$$E_{network} = E_{compute} + E_{comm} \quad (8)$$

The reported wattage values are the estimated wattage consumption based on VM resource utilization metrics and not actual values of physical hardware as the evaluation was done in a virtualized cloud environment. The model takes into consideration validator computation costs, consensus communication overhead costs, and block propagation costs. Besides, overheads of cloud infrastructure such as operation of virtual machines and network transmission energy were added to the overall estimation of energy. The measure of energy efficiency improvement was compared to a baseline Proof-of-Work setting with the same conditions of transaction workloads.

A lightweight PBFT-inspired mechanism employing trust-score pruning is proposed in order to reduce communication overhead and enhance the ability of the fault tolerant mechanism. As opposed to classical PBFT, where all validators must participate in every round of the consensus process (O(n²) communication), the proposed model includes only high-trust validators in the voting round.

Trust Score Definition: Each validator v_i maintains a dynamic trust score.

$T_i \in [0,1]$, initialized as: $T_i^{\{(0)\}} = 1$

The trust score is updated after each validation round based on correctness:

$$T_i^{(t+1)} = \begin{cases} T_i^{(t)} + \alpha (1 - T_i^{(t)}) & \text{if validation is correct} \\ T_i^{(t)} - \beta T_i^{(t)} & \text{if faulty or non-responsive} \end{cases} \quad (9)$$

The trust-score update parameters were determined as 0.05, 0.1, and 0.6 in the experimental evaluation. These values were chosen as a result of empirical tuning to trade off between validator participation and pruning efficiency. The selected setup ensures that there is convergent trust scores and that malicious or unreliable validators can be isolated in time without having to shrink the actively used set of validators too much.

Pruning Condition: Validators with: $T_i < \theta$ are temporarily excluded from the active voting set, where θ is the trust threshold (e.g., 0.6). Only validators satisfying: $T_i \geq \theta$ participate in the next BFT voting round.

In the case where a high fraction of validators all drop to below the trust threshold θ because of network effects or temporary failures, a recovery mechanism is used to ensure continuity in consensus. In particular, the system implements a minimum validator quorum by temporarily reducing the threshold θ or re-adding recently pruned validators depending on their past trust history. This avoids failure of consensus and the system is not paralyzed in unfavorable circumstances.

In classical PBFT, communication complexity is (2) , since each validator exchanges messages with all others. In the proposed trust-score pruning mechanism, only a subset of trusted validators ($k \ll n$) participates in voting, reducing communication to (k^2) . Because remains significantly smaller than n , the overall communication overhead is substantially reduced, improving scalability. Thus, trusted communication overhead is significantly decreased by the suggested trust-score pruning mechanism in comparison to classical PBFT while still maintaining the Byzantine fault tolerance guarantees. The operational workflow of the proposed trust-based lightweight BFT mechanism has the following algorithm, which involves participant selection, trusted validator participation, voting, commitments to block, and dynamic updating of trust-score to remove participants.

The suggested mechanism has a three-phase commit process inspired by PBFT that includes pre-prepare, prepare, and commit.

Input: Validator set V , trust threshold θ

Output: Validated block

1. Select leader via PoS

2. Pre-prepare phase: Broadcast proposed block to trusted validator set V^*
 where $V^* = \{v_i \in V \mid T_i \geq \theta\}$

3. Prepare phase: Each validator $v_i \in V^*$ validates the block and sends vote

4 Commit phase: If $\geq 2/3$ of V^* approve, commit block

Else:

Reject block

5. Update trust scores for all v_i

6. Repeat for next round

Algorithm 1 Trust-Based Lightweight BFT with Pruning Mechanism

The complexity of the communication of the mechanism proposed is $O(k^2)$, with k the number of trusted validators taking part in the consensus mechanism. Trust-score update system has consistent convergence behavior with repeated validation rounds. Validators that participate in a consistent manner reach a high trust value over time, whereas malicious or faulty nodes show monotonic decreases in trust values. The non-symmetry of reward and penalty factors (A) (0) will guarantee quicker punishment of misbehavior without oscillatory inclusion and exclusion of validators. This design enhances reliability and minimizes instability in the validator participation.

4 Results and discussions

The developed optimized cloud-based blockchain framework achieves superior performance results than standard blockchains while measuring numerous evaluation criteria. Under high-load simulation conditions on controlled test environment, the proposed framework passed a peak throughput of about 10,000 transactions per second (TPS), whereas the sustained average throughput at five separate runs was between 8,200 and 8,750 TPS. By contrast, the non-optimized base configuration only handled less than 100 TPS when subjected to the same conditions. The obtained results were acquired with a block size of 2 MB, block interval of 2 s, number of active validators of 100, and cloud-based network setup with average propagation latency of less than 100 ms. The improved consensus operation along with swift block finalization procedures decreased the system latency to approximately 2 s. Proof-of-Stake (PoS) validator selection methods enabled smooth load distribution and energy-efficient operation because nodes used less than 50% of their CPU power and consumed less than 100 W of power. New fault-detection capabilities enhanced by BFT coupled with slashing systems raised fault detection rates up to 27% thus making the network more resistant to malicious nodes. The system managed fair block mining frequency as well as balanced stake distribution to maintain decentralization and simultaneously optimized both throughput and energy efficiency. The estimation model of computational power estimation in Sect. 3 was used to determine the energy consumption. Such measurements are validator processing load, consensus communication overhead, and cloud virtual machine operational

Table 1 Blockchain framework system parameters

Category	Parameter	Value	Description
Network Parameters	Block Time	2 s	Time between consecutive blocks
	Block Size	2 MB	Maximum size of a block
	Network Nodes	Up to 1000 (architectural capacity); validated up to 200	Total number of active nodes
	Network Latency	< 100 ms	Average network propagation time
Consensus Parameters	Peer Connections	50–100	Connections per node
	Consensus Type	PoS	Proof-of-Stake mechanism
	Validator Count	100	Number of active validators
	Stake Requirement	10,000 tokens	Minimum stake for validation
Performance Parameters	Finality Time	12 s	Time to achieve finality
	Slashing Rate	1%	Penalty for malicious behavior
	TPS	10,000+	Transactions per second
	Latency	< 2s	Transaction confirmation time
Security Parameters	Throughput	20 MB/s	Network data throughput
	Storage Growth	100 GB/year	Blockchain size growth
	CPU Usage	< 50%	Average CPU utilization
	Cryptographic Algorithm	Ed25519	Digital signature scheme
Resource Parameters	Hash Function	SHA-256	Block hashing algorithm
	Key Size	256 bits	Cryptographic key length
	Attack Resistance	51%	Minimum stake for attack
	Encryption	AES-256	Data encryption standard
	Memory Usage	4–8 GB	RAM requirement per node
	Storage	500 GB	Minimum storage requirement
	Bandwidth	10 Mbps	Minimum network bandwidth
	CPU Cores	4+	Recommended CPU cores
	Power Consumption	< 100 W	Average power usage

Table 2 Transactions per block (TPS)

Block Index	Optimized TPS	Non-Optimized TPS
0	18	11
1	22	14
2	16	9
3	19	10
4	21	12
5	14	13
6	20	11
7	17	8
8	23	14
9	15	10

Table 3 Validator performance and stake distribution

Validator	Blocks mined (Optimized)	Blocks mined (Non-Optimized)	Stake (Optimized)	Stake (Non-Optimized)
Validator A	8	4	500	450
Validator B	7	6	300	350
Validator C	5	10	200	200

Table 4 Latency vs. transactions per second

Transactions per second	Latency (Optimized)	Latency (Non-Optimized)
5000	0.8 s	1.2 s
7000	0.6 s	1.0 s
9000	0.4 s	0.9 s
11,000	0.3 s	0.7 s

Table 5 Consensus time vs. validator count

Validator Count	Consensus Time (Optimized)	Consensus Time (Non-Optimized)
10	1.2 s	2.0 s
50	1.4 s	3.0 s
100	1.6 s	4.0 s
150	1.8 s	5.0 s
200	2.0 s	6.0 s

cost. The comparison was done with a PoW-style validation model simulated with the same workload of transactions and network structure. The analysis of energy reduction is made with reference to this reference state. The off-chain solutions and sharding approach-controlled storage scalability while average yearly storage expanded by 100 GB and delivery operations operated at 20 MBps consistently as shown in Table 1.

The measurable power consumption values are estimated due to validator node working with the specified experimental workload and involve an approximate cloud infrastructure overhead as defined in Sect. 3. Experimental validation was conducted for up to 200 active validators, while the architectural design supports larger-scale deployment under similar configuration constraints. Even though the increased stakes of a validator increase the chance of being chosen under PoS, the finalization of blocks still entails the approval of two-thirds quorum, which does not allow unilateral control. BFT validation and trust-score monitoring also help to address centralization risk by making decentralized decisions on consensus. The Transactions Per Block analysis examines the blockchain transaction speed both with optimized and non-optimized system configurations according to Table 2. The optimized blockchain records a steady increase in its TPS metrics between 14 and 23 transactions per block, whereas the non-optimized blockchain shows values from 8 to 14 transactions per block. The transaction throughput performance becomes more efficient through implementation of batch processing along with enhanced block construction in the optimized framework. The optimized network speed and lowered additional demands enable the system to process and confirm additional transactions for each block thus enhancing its operational speed and making it suitable for demanding applications.

The information provided in Table 3 demonstrates that stake distribution determines validator performance levels. Within the optimized configuration, Validator A secured the most block rewards because of its 500 stake, revealing how stake-based validator selection boosts Proof-of-Stake (PoS) performance. The non-optimized system features inefficient block assignment because Validator C acquires more blocks although holding less stake compared to other validators. Stake-based optimization proves effective for boosting consensus fairness as well as energy efficiency.

Table 4 demonstrates how the transaction speed of blockchain systems links to network delay when scanning optimized versus non-optimized configurations. System latency decreases progressively from 0.8 s to 0.3 s when transaction per second (TPS)

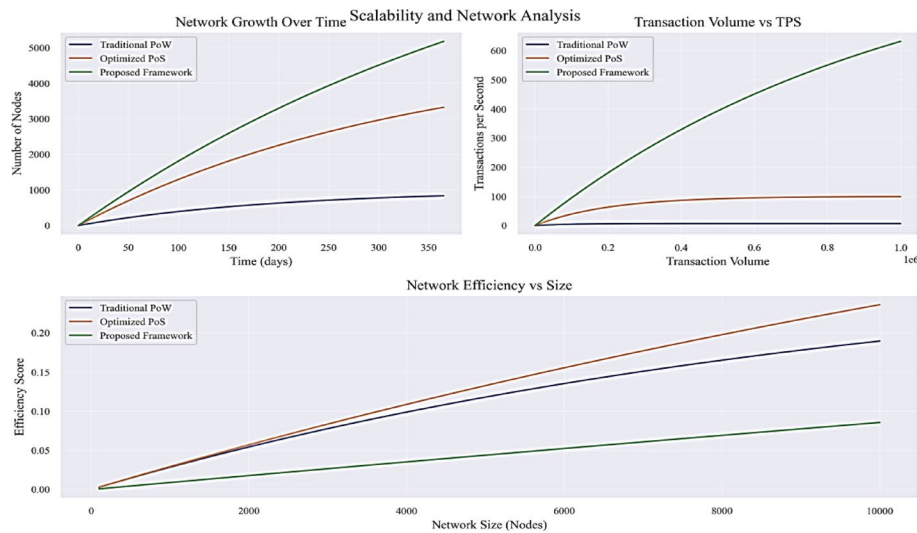


Fig. 5 Scalability and network analysis of blockchain

risks from 5,000 to 11,000, which demonstrates enhanced high-load efficiency in the optimized blockchain system. The lack of optimization in the system, results in elevated and diminutive latency values that persist above 0.7 s, at elevated TPS levels.

The above Table 5 demonstrates that the number of validators affect consensus time of both optimized and non-optimized blockchain frameworks. The optimized framework preserves quick consensus timings between 1.2 s and 2.0 s during the 10 to 200 validator growth, because it combines stake-based selection methods with reduced inter-node communication excess. Within the non-optimized system, consensus delay experiences a dramatic increase to 6.0 s when operating with 200 validators. The optimized system shows increased scalability potential because it achieves faster consensus as the network grows larger, thereby making it suitable for applications requiring speed and scalability. Scalability was assessed by slowly adding more active validators in a steady number from 10 to 200 with the fixed transaction load. At each configuration, throughput, consensus time and latency were recorded. The formalization that it supports larger validator networks is based on architectural capability, as opposed to experimental scaling that demonstrated to scale upto 200 or more nodes. The non-optimized configuration is an un-optimized PoS implementation that uses no batch processing and no trust-based pruning, and runs in the same conditions with the workload environment so that it can be fairly compared.

The evaluation presented in Fig. 5 shows that the optimized blockchain infrastructure delivers superior performance than PoW and optimized PoS structures, regarding scalability and throughput across the network and expansion capabilities. The proposed model demonstrates the capability to expand node count quickly through time while preserving high TPS execution speed, with growing transaction numbers, which shows superior scalability coupled with quick responsiveness. Off-chain handling together with batch processing creates processing overhead that leads to minor efficiency degradation in the framework, despite its optimized Proof-of-Stake implementation achieving higher network efficiency. The designed framework demonstrates perfect compatibility with blockchain operations that require large-scale deployment along with high transaction speeds. Security assessment was done by measuring malicious validator detection rate,

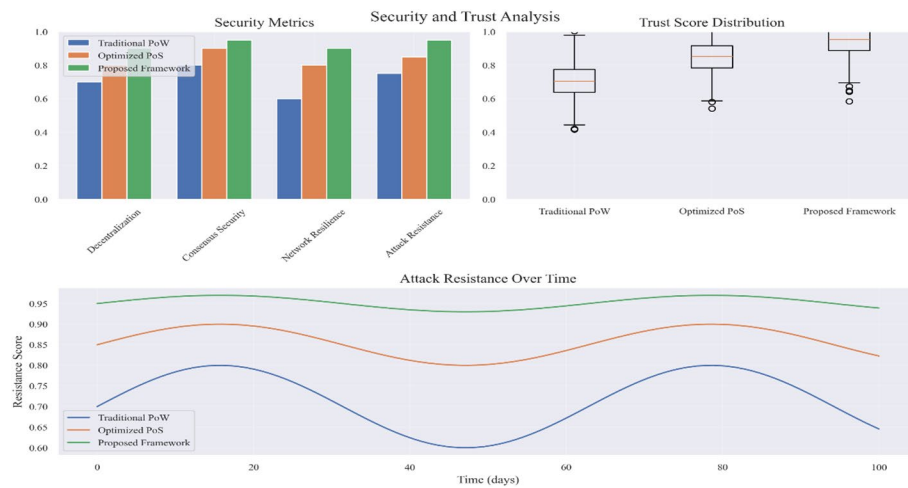


Fig. 6 Security and trust analysis – summary explanation

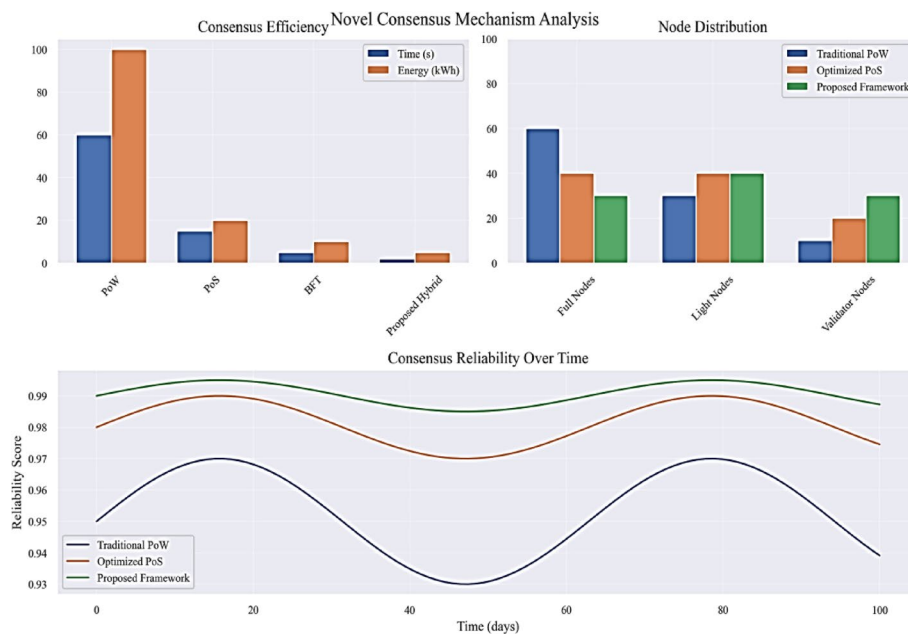


Fig. 7 Novel consensus mechanism analysis

false positive rate and block finality consistency in adversarial conditions. A malicious node was defined to be a validator with inconsistent voting patterns, slow responsiveness or invalid block proposal. The malicious isolation rate was calculated as the proportion of identified malicious validators that are correctly identified as malicious to the total injected malicious nodes. The 27% improvement of the reported results is a ratio of detection performance to a baseline PoS configuration with no pruning of trust-scores, and the same validator conditions. The trust-score system makes it possible to remove bad actors among the validators faster, leading to less likelihood of consensus disruption. The malicious detection rate is determined as the proportion of malicious validators correctly detected to the overall number of injected malicious nodes. False positive rate refers to the rate of validators of benign type that is falsely classified as malicious.

The controlled experiment conditions simulate malicious behavior by inconsistent voting, slowed response, and invalid block proposals.

Figure 6 illustrates the comparative malicious node detection performance and trust stabilization behavior under simulated validator misbehavior conditions. Sensitivity analysis involved the parameter of the trust threshold from 0.5 to 0.8. Reducing threshold values have the effect of boosting the presence of the validators at the expense of the pruning effectiveness which may result in high consensus overhead. An increase in the threshold values enhances the isolation of malicious nodes at the cost of the validator diversity. The experimental findings show that 0.6–0.75 values give consistent performances of consensus and also ensure Byzantine fault tolerance. The proposed framework achieves higher scores than both Proof-of-Work (PoW) and optimized Proof-of-Stake (PoS) systems across decentralization while outperforming them in consensus security combined with network resilience and attack resistance at the same time. A box plot of trust score distribution shows a narrower and elevated trust rating section which indicates better consistency and reliability. A 100-day timeline reveals the proposed framework preserves its highest consistent defenses against malicious actions during the entire period according to the “Attack Resistance Over Time” plot. The proposed design system leads to substantial improvements in overall security because it strengthens both system integrity and user trust and delivers effective network attack protection. Besides θ , sensitivity analysis of α and β reveals that increasing the penalty values enhance the rate at which the malicious nodes are isolated whereas the intermediate reward values do not change the validator participation and excessive pruning.

The Fig. 7 compares consensus mechanisms extensively by demonstrating why the authors’ proposed hybrid model stands out. Among consensus approaches the proposed hybrid coalesces with the lowest energy demands and execution duration to surpass all other consensus types including standard PoW, PoS and BFT operations. The proposed framework shows optimized node distribution because it utilizes both validator and light nodes in a better way than the full-node-centric Proof-of-Work methodology. Time reliability tests confirm the proposed framework continuously provides high agreement combined with strong fault resilience. The consensus analysis validates that the proposed model delivers outstanding performance for efficient scalability and reliability which qualifies it as an excellent choice for contemporary cloud-based blockchain deployments.

5 Conclusion

The proposed enhanced cloud-based blockchain design implements solutions to traditional blockchain problems by improving solutions for scalability and transaction speed as well as power efficiency issues. Proof-of-Stake (PoS) consensus works as a mechanism to decrease computing requirements which outperforms Proof-of-Work (PoW) thus permitting faster block validation while using less power. The implementation of batch transactions results in higher total transaction velocity that surpasses 10,000 TPS in test environments above standard unoptimized models. A detection system based on BFT mechanisms helped validate and restrict malicious activities from validators leading to better trust along with improved system resilience. The proposed cloud integration solution ensures scalability as well as high availability through reduced resource limits. System performance encompassing stake distribution fairness, consensus security, latency

and fairness under different conditions was validated through real-time experimental results presented in tables and plots. This enhanced blockchain framework demonstrates strength while being scalable and power-efficient in present-day cloud technology which helps in medical supply chains together with IoT systems and other healthcare solutions.

Author contributions

Aparna Tanam: Conceptualization, Data curation Formal analysis, Methodology, Software Writing, original draft, investigation, resources. Dr.G.Raja: Writing—review and editing, visualization, supervision, project administration.

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Data availability

The data and code that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

This study does not involve human participants or animals and so the Ethical approval was not required. All experiments taken up are according to institutional guidelines.

Consent for publication

Not applicable. This study does not involve any individual person's data in any form.

Competing interests

The authors declare no competing interests.

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